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# In-Situ Geo-Stress Fracturing Simulation Based on Geological and Engineering Integration Assists the Cost-Effective Development of Horizontal Wells

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## Abstract

Volume fracturing in horizontal wells has become the effective development method for tight conglomerate reservoir in MaHu oilfield, while its economic development has also been hindered due to the increasing cost of its large-scale implementation. Consequently, maintaining stimulation results while reducing the cost of fracturing operation has become the priority to realize the cost-effective development of this unconventional resource.

In this paper, a multi-aspect comprehensive research, combining 3D in-situ stress, rock mechanics analysis and laws of proppant migration and displacement, has been conducted to establish an integrated 3D model with geological and engineering attributes applicable for wells in several blocks of Mahu oilfield. The established model could be used to simulate the hydraulic fracture morphology, predict the fracturing effects of horizontal wells, 3D fracturing design optimization and production dynamic prediction. Based on the established model, series of pilot experiments, including energy-enhancing and energy-storing fracturing technology, adopting low-cost fracturing material combination, have been conducted in several blocks and showed encouraging results for cost-effective enhancement.

Typical application could be exemplified by numerically simulating the change of dynamic stress field for fractured wells, during which the variation characteristics of minimum principal stress, pore pressure and bottom-hole flow pressure during fracturing and production of horizontal wells could be revealed. Simulation results showed that the force exerted on the proppant increased from 17 MPa to 28 MPa during flowing and pumping period, which were lower than the strength of quartz sand, justifying the applicability for adopting quartz sand to substitute ceramic proppant for horizontal wells in the area. Several horizontal wells in the pilot area of MaHu oilfield, fractured with quartz sand and ceramic proppant respectively, turned out the average oil production rate of horizontal sections using quartz sand and ceramic proppant was 3.28 ton/m and 3.68 ton/m respectively. While the cost of proppant has dropped to almost one-third of that of

previous counterpart yet there was no significant difference in the average oil production rate. This progress facilitated the large-scale adoption of quartz sand in volume fracturing to reduce costs.

The development and application of proposed 3D model for fracturing simulation, which poses guiding significance for well placement, fracturing and production design and optimization, underscores the advantage of digital driven technique in cost-effective enhancement for unconventional resource development.

## Introduction

The massive tight conglomerate oil field, discovered in the Mahu Depression of the Junggar Basin, has become one of the new frontiers in China's onshore unconventional hydrocarbon resource. The oil resource potential is estimated to 46.7 billion tons with the proven geological reserves reaching 5.2 billion tons. The tight oil reservoir is buried at a depth of 2500–4000 meters with the main oil layers distributed in the Permian and the Triassic strata. The reservoir thickness ranges from 0.5 to 50 meters with effective porosity ranging from 7% to 12% and saturation ranging from 45%' to 65% (Tian et al., 2023). Despite challenges including but not limited to dense and hard conglomerate, poor reservoir physical properties and strong heterogeneity, horizontal wells and volume fracturing have become the effective method for development of this unconventional oil resource (Tian et al., 2023; Xu et al., 2019). This is crucial for Xinjiang Oilfield to achieve sustainable hydrocarbon production as it's increasing difficult to maintain economic output in those mature fields.

While the increasing cost of large-scale volume fracturing implementation has hindered the cost-effective development of this unconventional resource, effective cost-saving measures are urgently needed to reduce fracturing operation expenses while ensuring oilfield production consequently. According to statistics, fracturing material costs, primarily proppant and fracturing fluid, account for over 40% of reservoir stimulation operation expenses. Proppant used in fracturing are mainly divided into quartz sand and ceramic proppant, with a price difference of 2000–2600 yuan per cubic meters. The difference in reservoir stimulation costs when using quartz sand and ceramic as proppant ranges from 3.5 to 5 million yuan for a single horizontal well.

Conducting research for using quartz sand as a substitute for ceramic proppant is crucial for achieving cost reduction. However, before using quartz sand to replace ceramic proppant, it is necessary to determine whether such replacement is feasible and how it should be carried out. To this end, this paper presents the technical research conducted on using quartz sand to replace ceramic proppant, aimed at determining the required fracture conductivity, the actual stress state of quartz sand in the formation, and the conductivity provided by proppant under actual formation stress conditions. In this process, a comprehensive 3D numerical model, used for reservoir production and dynamic stress simulation based on the geological and engineering integration, has been established for fractured wells. Additionally, laboratory tests were conducted to determine the influence of closure stress, particle size and concentration of quartz sand on the conductivity under actual down-hole conditions. Indoor tests were also conducted to find out the sedimentation and transport characteristics of quartz sand in both slick water and cross-linked gel, which helped to develop the suitable design in fracturing operation. Based on the research findings, pilot tests were conducted to substitute ceramic proppant with quartz sand, confirming that in a certain block of the Mahu Oilfield, quartz sand could successfully replace ceramic proppant without significantly sacrificing oil production rate, thereby significantly reducing the development costs of this tight conglomerate oil reservoir.

This finding has promoted the large-scale replacement of ceramic proppant with quartz sand in the Mahu Oilfield. It has also confirmed the positive role played by this digital-driven method, namely the established 3D model based on geological and engineering integration, in the process of reducing costs and increasing efficiency in the exploitation of this unconventional resource.

## Challenges for substituting ceramic with quartz sand

In fracturing operation, increasing the amount of proppant injected typically improves the conductivity of reservoir fractures. However, from an economic feasibility perspective, it is necessary to consider not only the absolute conductivity, but also the economic efficiency of the ratio between the amount of proppant injected and the hydrocarbon production output, namely the input-output ratio. Differ from sandstone or shale reservoirs, the roughness of fracture surfaces in the Mahu tight conglomerate reservoir has a significant effect on the migration of sand-carrying fluid. In areas with larger openings, the sand-carrying fluid advances in a finger-like manner. Rough fracture surfaces restrict the effective placement of proppant along the length of fractures, thereby affecting its conductivity (Liu et al., 2024; Li et al., 2025;Li et al., 2025). Additionally, the closure stress of this tight conglomerate reservoir generally exceeds 30 MPa, and an increase in closure stress during fracturing operation reduces the effective conductivity. Furthermore, uneven stress distribution on proppant in gravel zones, combined with high closure stress and stress concentration, leads to a high rate of breakage of quartz sand, thereby reducing the overall effective conductivity. In this context, replacing quartz sand with ceramic proppant not only requires clarifying the fracture conductivity required for achieving economic production capacity, but also determining whether quartz sand can effectively support fractures during different production stages. Additionally, it is crucial to understand the influence of changes in formation closure stress, particle size and concentration of quartz sand on fracture conductivity, as well as the migration and sedimentation laws of quartz sand in different hydraulic fluids during actual production. These findings could help to optimize operational design, which is essential for maintaining the required conductivity during long-term production.

## Research conducted for substituting ceramic proppant with quartz sand

Research relating to the integration of geology and engineering has been widely applied to optimize drilling, prevent and control down-hole complexity and optimize or enhance production in the Mahu Oilfield. A better understanding of geology could help to clarify the difficulties of fracturing. Numerous novel fracturing technologies, including energy-enhancing and energy-storing fracturing, low-cost proppant material combination and substitution, have been developed and applied in several pilot areas based on research findings relating to 3D in-situ geo-stress modelling, rock mechanics analysis, proppant migration and sedimentation laws (Chen et al., 2024; Lu et al., 2024; WANG et al., 2024; Li et al., 2019; ZHANG et al., 2023; Qu et al., 2022; Xiang et al., 2022; Cai et al., 2021; Yang et al., 2023; Jia et al., 2025; Fan et al., 2023) and etc. The use of low-cost fracturing material substitution is a prime example of how geological and engineering integration technologies can support the cost-effective enhancement in this unconventional hydrocarbon resource.

## Simulation and evaluation of required fracture conductivity

Based on multiple sources of data including but not limited to well logging, seismic surveys, core laboratory tests, dynamic production, and in-situ micro-seismic measurements, a coupled fluid-solid finite element computational model was developed to establish an integrated 3D model of the Mahu tight oil reservoir which incorporates both geological and engineering attributes. The model encompasses modules for 3D structural and geological modelling, rock mechanics analysis and in-situ geo-stress modelling, fracturing and reservoir numerical simulation. This enables the multi-dimensional visualization of the characteristics of this tight oil reservoir, the simulation of fracture morphology under specified operating parameters, the tracking of temporal and spatial changes in in-situ stress before and after fracturing, the evaluation of the fracture conductivity, and the prediction of the production rate for fractured wells. Based on the established model, a method was developed, using economically feasible cumulative recovery rate as the objective function for matching the required fracture conductivity, in order to achieve the desired reservoir permeability as shown in Fig. 1. This method clarified the required fracture conductivity, which ranged from 15 to 40 D·cm, for horizontal wells with cluster spacing ranging 20–35 m in the pilot area.

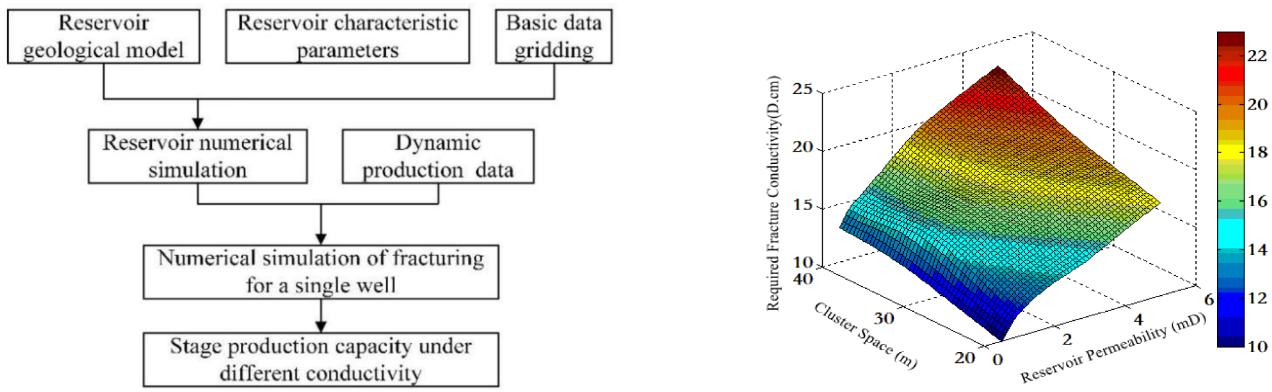


Figure 1—Flowchart (left) and results (right) of numerical simulating for required conductivity

## Prediction of effective stress exerted on proppant

The effective stress exerted on the proppant in the wellbore is primarily influenced by fracture closure stress and changes in the bottom-hole fluid pressure. Significant fluctuations in bottom-hole pressure could be observed before and after fracturing, as well as during flowing or pumping stage, which in turn cause significant changes in the stress exerted on the proppant. By constructing a dynamic model of stress field variation in fractured wells, it is possible to simulate bottom-hole fluid pressure and the stress variations experienced by proppant within reservoir fractures under actual operating conditions. Since the fluid within the fracture and the proppant jointly bear the closure stress generated by fracture deformation, the simulation of the test wells revealed that the closure stress before and after fracturing exhibited a 'rise, recovery, and decrease' pattern in response to changes in formation pore pressure, as shown in Figure 2. During the natural flowing and pumping period, the closure stress experienced further decrease. Simulation results showed that the effective stress exerted on the proppant during the natural flowing period was 7–17 MPa, and 23–28 MPa during the pumping period. These values are lower than the strength of quartz sand, which justifies the engineering feasibility of using quartz sand as a substitute for ceramic proppant during fracturing operation in horizontal wells in the pilot area.

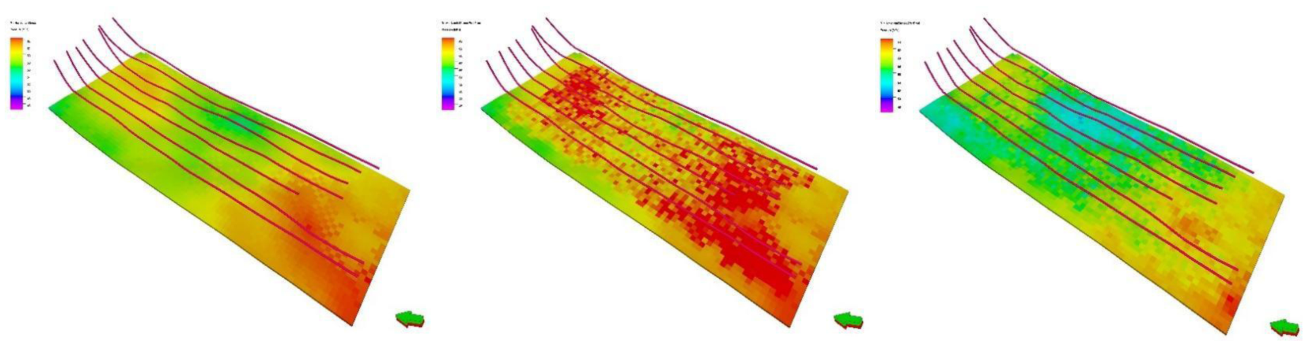

Figure 2—Simulated minimum horizontal principal stress Hmin:
before fracturing(left), post fracturing(middle,) after production (right)

## Long term conductivity determination based on indoor experiments

The conductivity of fractures is primarily determined by fracture width and the amount of proppant packed per unit area, without considering the settlement and fragmentation of proppant during fracture closure. While changes in closure stress significantly influence fracture conductivity during the fracturing and production phases. Moreover, stress concentration in conglomerate rock may lead to quartz sand fragmentation, and factors such as the particle size and concentration of the quartz sand also influence fracture conductivity. To address this issue, indoor laboratory tests were conducted on fracture conductivity using quartz sand in the conglomerate rock slab. Scanning electron microscopy was also used after the tests, and it was determined that the primary factor affecting conductivity was stress concentration which may lead to sand fragmentation. The study also revealed that quartz sand with smaller particles exhibited greater resistance to fragmentation. When the fracture width is fixed, using quartz sand with smaller particles allows for more layers to be packed into the fracture, thereby reducing the stress concentration effect.

**Figure 3—Laboratory tests for fracture conductivity with quartz sand in conglomerate rock slab**

Figure 4 shows the experimental conductivity when using quartz sand with different particle sizes and concentrations under different closure stresses. As can be seen in the figure, the conductivity decreases as the closure stress increases, and increasing the sand concentration or decreasing particle size could help to improve conductivity. Based on the experimental data, a model was established for fracture conductivity prediction under actual down-hole conditions which considered proppant fragmentation, particle size, sand concentration and closure pressure. The model's predictive results showed a high degree of agreement with the experimental data. Considering the increase in closure stress could potentially lead to an reduction in long-term conductivity as production progresses after fracturing, it is also necessary to consider the influence of increase in stress exerted on proppant over conductivity during the natural flow and pumping stage, in order to select suitable operating parameters that will maintain the required long-term conductivity. Simulation showed that the effective closure stress for horizontal wells in the pilot area could reach up to 45 MPa during the pumping period, and the fracture conductivity could still be maintained at 20–40 D·cm using 20/40 mesh quartz sand with a sand concentration of 8 kg/m³ as demonstrated by the experimental findings. This met the required fracture conductivity for achieving economic hydrocarbon production in this tight conglomerate reservoir.

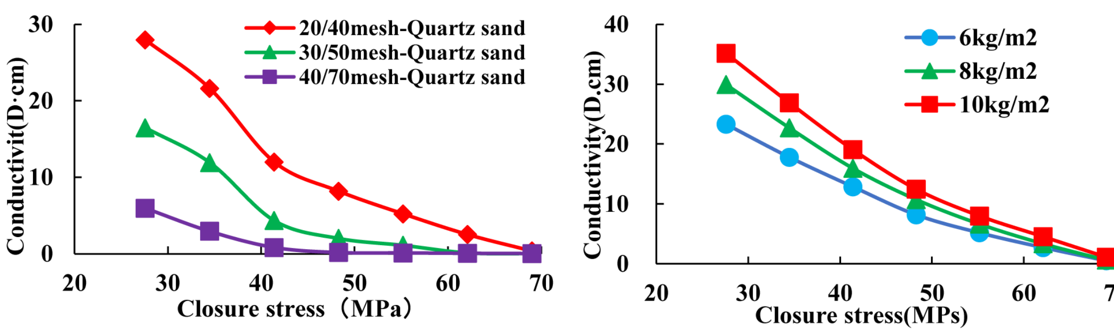


**Figure 4—Experimental conductivity using quartz sand as proppant: 8kg/ m² for different particle size(left); 20/40 mesh with different concentration**

Additionally, laboratory tests were also conducted on the sedimentation and migration characteristics of quartz sand with different particle sizes in both slick-water and cross-liked gels, as shown in Figure 5. These tests revealed that quartz sand, with a smaller particle size and density, exhibited a slower settling rate and a longer migration distance. Based on this data, a dynamic settling formula for quartz sand was developed that takes into account the roughness of the fracture surface in this dense, tight conglomerate reservoir. Based on the results of the indoor experiments and the developed formula, it was recommended that slick-water proppant slugs be used to ensure a high sand concentration in the near-wellbore region, while a high-sand-ratio gel fracturing method should be adopted to ensure more effective proppant transport of quartz sand to the far-field fractures. This operation design for using different fracturing fluids for different wellbore sections could help to improve the overall conductivity of fractured horizontal wells.

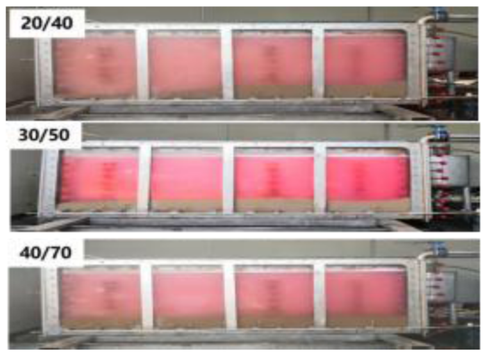


**Figure 5—Laboratory tests on sedimentation and transport of quartz sand**

## Filed application and results

Based on simulation results and experimental findings, quartz sand was used to replace ceramic proppant in fracturing operation of horizontal wells in a certain pilot area of the Mahu Oilfield. This area has a reservoir depth of 2568m–3260m and a formation pressure coefficient of 1.11–1.18. Natural flow stopped when the wellhead pressure dropped to 0.5 MPa after fracturing. The maximum effective closure stress during natural flow was found to be around 28.6 MPa in the simulation analysis. Two adjacent horizontal wells (A and B) were fractured, with quartz sand used in well A and ceramic proppant in well B. The relevant information and fracturing operation parameters are shown in Tab 1.

**Table 1—Information and fracturing parameters for horizontal well A and B**

| Well Name | A | B |
|---|---|---|
| Vertical Depth(m) | 3106 | 3103 |
| Horizontal Length (m) | 1612 | 1692 |
| Stage Count /Cluster Count | 21/41 | 22/42 |
| Reservoir Contact Ratio(%) | 91.7 | 85.7 |
| Proppant Type | Quartz sand | Ceramic |
| Proppant Volume($m^3$) | 1814 | 1597 |
| Proppant Concentration($m^3$/m) | 1.13 | 0.95 |
| Fracturing Fluid Volume($m^3$) | 25785 | 24578 |

The reservoir properties of the horizontal sections of two wells were essentially same, while Tab 1 shows that the fracturing operation parameters of two wells were almost identical, with only a slight difference in proppant concentration. Fig. 6 shows the daily and cumulative oil production of two wells within 600 days after fracturing. Statistics show that, 587 days after the two wells were put into production, the cumulative oil production for well A and B were 11743 and 11511 cubic meters, respectively.

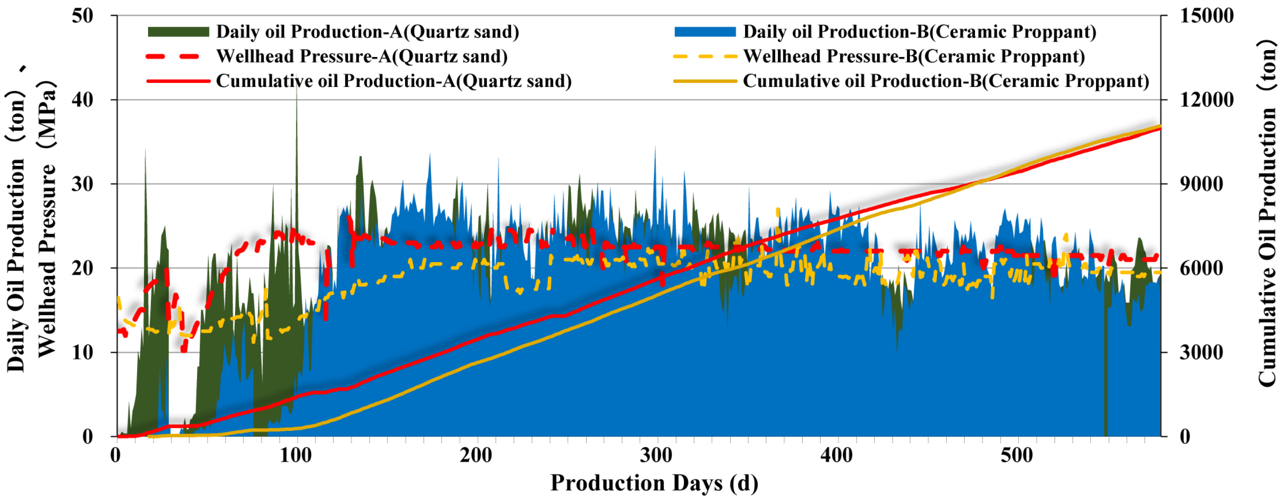


**Figure 6—Daily and cumulative oil production as well as wellhead pressure curves for wells A and B after fracturing**

Fig. 7 shows the cumulative oil production of eight horizontal wells in the pilot area. These wells are located in reservoirs with similar physical characteristics, and there was no significant difference in fracturing operation parameters. Of these eight wells, four wells were fractured with quartz sand and another four were fractured with ceramic proppant. After 270 days of production, a difference of only 1.5 tons in the average daily oil production rate was observed: 5024 tons for wells using quartz sand and 5,429.4 tons for wells using ceramic proppant. Considering the length of the horizontal section, the cumulative oil production was converted to cumulative oil production per meter. The average oil production per meter for horizontal wells using quartz sand as proppant was 3.28 tons/m and for horizontal wells using ceramic proppant it was 3.68 tons/m. This demonstrated that there was no significant difference in the average oil production per meter for horizontal wells using different proppant in the pilot area. Given that the cost of ceramic proppant is nearly 2–3 times higher than that of quartz sand, the pilot area effectively reduced fracturing expenses without significantly sacrificing oil production by replacing ceramic proppant with quartz sand.

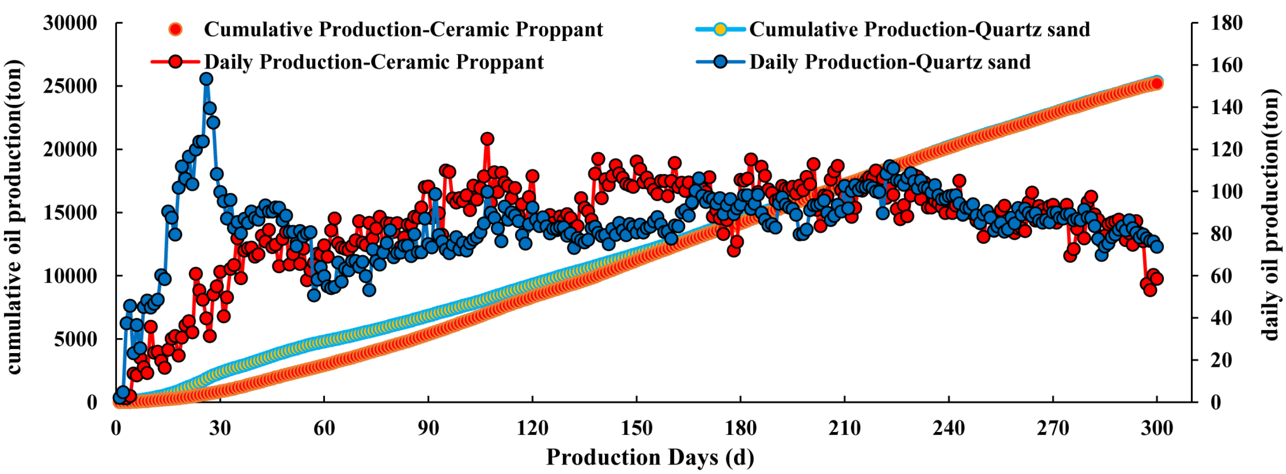


**Figure 7—Daily and cumulative oil production of eight wells fractured using quartz sand and ceramic proppant**

## Conclusion

In order to replace ceramic proppant with quartz sand in fracturing operation so as to reduce the development costs of the Mahu tight conglomerate oil reservoir, an integrated geological and engineering 3D model was developed alongside laboratory tests to determine the factors influencing the conductivity, migration and sedimentation patterns of quartz sand. Based on the analysis of numerical modelling and findings of laboratory tests, tests were conducted in the pilot areas to replace ceramic proppant with quartz sand. The progress achieved can be illustrated as follows:

1. A comprehensive, three-dimensional numerical model incorporating geological and engineering attributes was developed to simulate the production capacity of horizontal fractured wells under specified conditions, such as the number of wells, well spacing and cluster spacing. Based on the established model, the required fracture conductivity for economically viable hydrocarbon production in this tight reservoir could be obtained.
2. The developed model can also simulate changes in formation closure stress, down-hole fluid pressure and the stress exerted on the proppant before, during and after fracturing, as well as during different phases of production. This is essential for determining the boundary conditions necessary for replacing ceramic proppant with quartz sand in the reservoir.
3. Laboratory tests were also conducted and revealed the effects of particle size, concentration of quartz sand and formation closure stress on reservoir fracture conductivity, as well as the migration and settling patterns of quartz sand in both slick-water and cross-linked gel. These findings are crucial for developing appropriate fracturing design to ensure that reservoir fractures can maintain the required conductivity during the long-term production.
4. Based on previous analysis, the production data from test horizontal wells in the pilot area confirmed the both economic and engineering feasibility of replacing ceramic proppant with quartz sand. This technology could effectively reduce fracturing expenses without significantly sacrificing the oil production rate.
5. Pilot tests involving the replacement of 100% ceramic proppant with various proportions of quartz sand and ceramic proppant are also underway in reservoirs with greater burial depth and higher closure stress. The aim is to expand the scope of application of low-cost proppant substitution, so as to further reduce the development expenses.

This integrated geological and engineering model can be used to optimize the 3D deployment of horizontal wells, fracturing designs, and production predictions. It provides guidance on the cost-effective development of this tight conglomerate reservoir and confirms the effectiveness of this digital-driven technique in improving the cost-effectiveness of unconventional hydrocarbon resource development. Furthermore, the methodology and process presented in this paper can serve as a reference for the profitable development of other similar unconventional reservoirs.

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